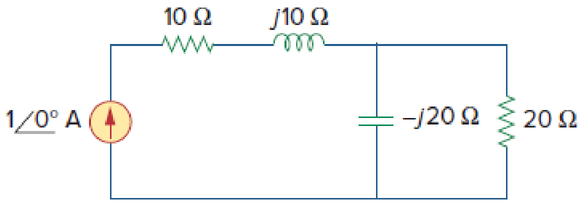
Chapter 11, Problem 54P

Problem

For the network in Fig. 11.73, find the complex power absorbed by each element.

Figure 11.73:



Step-by-step solution

Step 1 of 10

Refer to Figure 11.73 in the text book.

The current flowing through the $10\ \Omega$ resistor and $j10\ \Omega$ inductor is same as the value of the current source. That is, $1\angle 0°\ \text{A}$.

The current $1\angle 0°\ \text{A}$ divides between the $-j20\ \Omega$ capacitor and $20\ \Omega$ resistor.

Calculate the current I_{-j20} passing through the capacitor using current division rule.

$$\begin{aligned} I_{-j20} &= (1\angle 0°)\left(\frac{20}{20-j20}\right) \\ &= (1\angle 0°)(0.5+j0.5) \\ &= (1\angle 0°)(0.707\angle 45°) \\ &= 0.707\angle 45°\ \text{A} \end{aligned}$$

Therefore, the value of the current I_{-j20} passing through the capacitor is $0.707\angle 45°\ \text{A}$.

Step 2 of 10

Calculate the value of the current I_{20} passing through the $20\ \Omega$ resistor.

$$\begin{aligned} I_{20} &= (1\angle 0°)\left(\frac{-j20}{20-j20}\right) \\ &= (1\angle 0°)(0.5-j0.5) \\ &= (1\angle 0°)(0.707\angle -45°) \\ &= 0.707\angle -45°\ \text{A} \end{aligned}$$

Therefore, the value of the current I_{20} passing through the resistor is $0.707\angle -45°\ \text{A}$.

Step 3 of 10

Calculate the voltage across the $10\ \Omega$ resistor.

$$\begin{aligned} V_{10} &= (1\angle 0°)(10) \\ &= 10\angle 0°\ \text{V} \end{aligned}$$

Therefore, the voltage V_{10} across the $10\ \Omega$ resistor is $10\angle 0°\ \text{V}$.

Step 4 of 10

Calculate the complex power S_{10} absorbed by the $10\ \Omega$ resistor.

$$S_{10} = \frac{1}{2} V_{10} I_{10}^{*}$$

Substitute $1\angle 0°\ \text{A}$ for I_{10} and $10\angle 0°\ \text{V}$ for V_{10}.

$$\begin{aligned} S_{10} &= \frac{1}{2}(10\angle 0°)(1\angle 0°)^{*} \\ &= (5\angle 0°)(1\angle 0°) \\ &= 5\angle 0°\ \text{VA} \end{aligned}$$

Therefore, the value of the complex power S_{10} is $\boxed{5\angle 0°\ \text{VA}}$.

Step 5 of 10

Calculate the voltage across the $j10\ \Omega$ inductor.

$$\begin{aligned} V_{j10} &= (1\angle 0°)(j10) \\ &= (1\angle 0°)(10\angle 90°) \\ &= 10\angle 90°\ \text{V} \end{aligned}$$

Therefore, the voltage V_{j10} across the $j10\ \Omega$ resistor is $10\angle 90°\ \text{V}$.

Step 6 of 10

Calculate the complex power S_{j10} absorbed by the $j10\ \Omega$ inductor.

$$S_{j10} = \frac{1}{2} V_{j10} I_{j10}^{*}$$

Substitute $1\angle 0°\ \text{A}$ for I_{j10} and $10\angle 90°\ \text{V}$ for V_{j10}.

$$\begin{aligned} S_{j10} &= \frac{1}{2}(10\angle 90°)(1\angle 0°)^{*} \\ &= (5\angle 90°)(1\angle 0°) \\ &= 5\angle 90°\ \text{VA} \end{aligned}$$

Therefore, the value of the complex power S_{j10} is $\boxed{5\angle 90°\ \text{VA}}$.

Step 7 of 10

Calculate the voltage across the $20\ \Omega$ resistor.

$$V_{20} = I_{20}(20)$$

Substitute $0.707\angle -45°\ \text{A}$ for I_{20}.

$$\begin{aligned} V_{20} &= (0.707\angle -45°)(20) \\ &= 14.14\angle -45°\ \text{V} \end{aligned}$$

Therefore, the voltage V_{20} across the $20\ \Omega$ resistor is $14.14\angle -45°\ \text{V}$.

Step 8 of 10

Calculate the complex power S_{20} absorbed by the $20\ \Omega$ resistor.

$$S_{20} = \frac{1}{2} V_{20} I_{20}^*$$

Substitute $0.707\angle -45°$ for I_{20} and $14.14\angle -45°\ V$ for V_{20}.

$$\begin{aligned} S_{20} &= \frac{1}{2}(14.14\angle -45°)(0.707\angle -45)^* \\ &= (7.07\angle -45°)(0.707\angle 45°) \\ &= 5\angle 0°\ VA \end{aligned}$$

Therefore, the value of the complex power S_{20} is $\boxed{5\angle 0°\ VA}$.

Step 9 of 10

Calculate the voltage across the $-j20\ \Omega$ capacitor.

$$V_{-j20} = I_{-j20}(-j20)$$

Substitute $0.707\angle 45°\ A$ for I_{-j20}.

$$\begin{aligned} V_{-j20} &= (0.707\angle 45°)(20\angle -90°) \\ &= 14.14\angle -45°\ V \end{aligned}$$

Therefore, the voltage V_{-j20} across the $-j20\ \Omega$ capacitor is $14.14\angle -45°\ V$.

Step 10 of 10

Calculate the complex power S_{-j20} absorbed by the $-j20\ \Omega$ capacitor.

$$S_{-j20} = \frac{1}{2} V_{-j20} I_{-j20}^*$$

Substitute $0.707\angle 45°$ for I_{-j20} and $14.14\angle -45°\ V$ for V_{-j20}.

$$\begin{aligned} S_{-j20} &= \frac{1}{2}(14.14\angle -45°)(0.707\angle 45)^* \\ &= (7.07\angle -45°)(0.707\angle -45°) \\ &= 5\angle -90°\ VA \end{aligned}$$

Therefore, the value of the complex power S_{-j20} is $\boxed{5\angle -90°\ VA}$.